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27 March 1968
Superseding
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21 October 1965

MILITARY SPECIFICATION

CARTRIDGE, DELAY, MARK 87 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 87 MOD 0 delay cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-116	Preservation, Methods of
MIL-T-23132	Tungsten Delay Composition
MIL-P-20444	Primer, M42
MIL-P-22264	Powders, Ignition, Gasless

FSC 1377

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STANDARDS

Military

MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Bureau of Naval Weapons

DL 2518218	Cartridge, Delay, MARK 87 MOD 0
LD 537902	Container, Inner, for Cartridges, Power Source
LD 537903	Container, Outer, for Cartridges, Power Source

Naval Air Systems Command

DL 2838183	Test Set for Cartridge, Delay (.38 Cal. Size)

PUBLICATIONS

BUSANDA Instruction 4030.17	Preservation, Packaging, Packing and Marking; Navy Policy for

Bureau of Naval Weapons

OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43	Preparation of Quality Assurance Provisions

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 87 MOD 0 delay cartridge covered by this specification shall be manufactured in accordance with DL 2518218 and all documents listed thereon.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective Bureau of Naval Weapons drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 88 cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 to determine that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification, the M42 primer (MIL-P-20444), the delay composition (MIL-T-23132), and the gasless ignition powder (MIL-P-22264) are defined as primary components. See 4.3.3.

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3.4 Primer. The primer shall be furnished from lots manufactured within a year past, loaded with Olin Mathieson Chemical Corporation Primer Mixture Number 793 and accepted in accordance with MIL-P-20444.

3.4.1 X-ray. All primers shall be X-rayed and screened for defects as outlined in 4.7.1.

3.5 Tungsten delay composition. The tungsten delay composition shall meet all requirements of MIL-T-23132.

3.6 Gasless ignition powder. The gasless ignition powder shall be in accordance with the provisions of MIL-P-22264, Composition A-1A.

3.7 Propellant charges.

3.7.1 Quantity. All propellant charges shall be as specified on Bureau of Naval Weapons Drawing 2518218 of DL 2518218. Charges shall be determined by weight rather than by volume measurements.

3.7.2 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.7.3 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.2

3.8 Cartridge inspection.

3.8.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.8.1.

3.8.2 Leakage. There shall be no leakage of air in the completed cartridge. See 4.8.2.

3.8.3 X-ray. All cartridges shall pass the X-ray examination of 4.8.3.

3.9 Functional and ballistic requirements. Cartridges from each production lot, selected in accordance with 4.8.4, shall meet the requirements listed below when subjected to the test prescribed in 4.8.4.

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3.9.1 Delay time. Delay time is defined as the time from actuation of the firing mechanism to start of pressure rise. The lower and upper limits of delay time over the temperature range, -10°F to +135°F, shall be 1.5 and 3.5 seconds, respectively. See 4.9.1.

3.9.2 Maximum pressure. The delay cartridge shall deliver a maximum pressure of not less than 4050 psia, when tested in accordance with 4.8.4.

3.9.3 Misfire. There shall be no misfires. See 4.9.3.

3.10 Environmental tests. The first five production lots from a single supplier shall pass the environmental test of 4.8.5 after which cartridges from each fifth production lot shall be subjected to these tests. If any lot fails to pass these tests using the acceptability criteria of 4.9, they shall be required on every production lot until five consecutive production lots have passed.

3.11 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with provisions of MIL-STD-414. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation and size shall be in accordance with MIL-STD-414.

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4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 88 cartridges shall be prepared. Seventy-eight of these cartridges shall be expended in the tests listed below and ten shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.8.3 of this specification. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625, except as noted otherwise: leak test (78); X-ray (78); 40-foot drop (6); 15 "g" shock (12); vibration (12); temperature and humidity cycling, as modified by paragraph 4.8.5.1 of this document (12). In addition, 12 untreated cartridges each shall be conditioned at -15°F +0°F -5°F, +70°F ± 2°F and 130°F +5°F -0°F. The number in parentheses denotes the number of cartridges allocated for test. Firings of all cartridges, except those from the 40-foot drop test, shall be conducted in the test fixture of 4.8.4.1. Each cartridge fired from the preproduction sample shall have a delay time within the range 1.5 seconds to 3.5 seconds and a maximum pressure no less than 4050 psia. Cartridges which fail to pass the leak test and X-ray examination shall be replaced by the contractor at no additional cost to the Government. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. The suggested production lot size is 301 minimum and 1300 maximum cartridges. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the functional and ballistic tests shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide himself with gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification

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symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 2518218 and the following tests and examinations.

Item	Nature of Test	Requirement	Test	Classification (WR-43)
Primary Component	Single lot verification	3.3	4.3.3	(M101)
Primer	Age, primer mixture, and acceptance verification	3.4	---	(C1)
Primer	X-ray	3.4.1	4.7.1	(C2)
Delay Composition	Acceptance verification	3.5	---	(C3)
Ignition Powder	Acceptance verification	3.6	---	(C4)
Propellant	Acceptance and weight verification	3.7.1	---	(C5)
Smokeless Powder	Condition	3.7.3	4.7.2	(C6)
Cartridge	Visual inspection	3.8.1	4.8.1	(M102)
Cartridge	Leakage	3.8.2	4.8.2	(C7)
Cartridge	X-ray	3.8.3	4.8.3	(C8)
Cartridge	Functional and ballistic tests	3.9	4.8.4, 4.9	(C9)
Cartridge	Environmental tests	3.10	4.8.5, 4.9	(C10)

4.6.1 Packing, packaging and marking. (M103) The inspector shall ascertain that the packing, packaging, and the container markings conform to Section 5 of this specification.

4.7 Tests.

4.7.1 Primer X-ray. Primers should be mounted in suitable grooved racks. For the most revealing exposure, the primers are laid on their side with the anvil cutouts vertical. Pressure-sensitive tape may be used as an aid in positioning the primers. Reject any that are questionable from any standpoint, especially for missing or abnormally thin charge, charge broken, cracked, or porous; misplaced or cracked anvil or other improperly positioned or deformed inert parts.

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4.7.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable portions of paragraphs 1308-1318 and 1322 of OP 5. Evidence of decomposition shall be cause for rejection.

4.8 Cartridge inspection and tests.

4.8.1 Visual inspection. All cartridges produced shall be subjected to visual inspection. Reject for burrs, split or cracked lip in the crimped area, damage to closure disc, bulged or cracked case, or any other defect which may conceivably prevent entry into the firing chamber, or affect stability or resistance to moisture.

4.8.2 Leakage. Each cartridge in the preproduction sample or ballistic sample will be leak tested in a dry-gas leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc/sec air at a pressure differential of 1 atmosphere ± 0.1 atmosphere shall be considered defective. If one or more cartridges of the ballistic sample leak, then a leak test of each cartridge purchased under contract for the lot being tested shall be required. If a 100 percent leak test of the lot is required, it shall be at the expense of the contractor. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.8.3 X-ray examination. Examine all cartridges by X-ray and any observable imperfections in assembly shall be cause for rejection. X-ray plates may be destroyed after ballistic acceptance of cartridges.

4.8.4 Functional and ballistic tests.

4.8.4.1 Test fixture. The test fixture shall be in accordance with Naval Air Systems Command DL 2838183. A record of delay time and pressure versus time shall be taken for each firing.

4.8.4.2 Low temperature. Cartridges selected from the production lot in accordance with Table A2, Inspection Level V, of MIL-STD-414 shall be inserted in sleeves (NAVAIR Drawing 2838190 of DL 2838183) and conditioned to a temperature of -15°F to -10°F for a period of at least 12 hours (to establish thermal equilibrium) and fired in the test fixture within 2 minutes after removal from the conditioning chamber.

4.8.4.3 Normal temperature. Cartridges selected from the production lot as in 4.8.4.2 shall be inserted in sleeves (NAVAIR Drawing 2838190 of DL 2838183) and conditioned at a temperature of +70°F ± 2°F for a period of at least 12 hours (to establish thermal equilibrium) and fired in the test fixture within 2 minutes after removal from the conditioning chamber.

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4.8.4.4 High temperature. Cartridges selected from the production lot as in 4.8.4.2 shall be inserted in sleeves (NAVAIR Drawing 2838190 of DL 2838183) and conditioned at a temperature of +130°F to +135°F for a period of at least 12 hours (to establish thermal equilibrium) and fired in the test fixture within 2 minutes after removal from the conditioning chamber.

4.8.4.5 Additional cartridges. In addition to the sample of cartridges selected from the production lot for the above tests, ten additional cartridges shall be furnished for investigative purposes.

4.8.5 Environmental tests.

4.8.5.1 Temperature and humidity cycling. Cartridges selected from the production lot in accordance with Table A2, Inspection Level III of MIL-STD-414 shall be subjected to temperature and humidity cycling for seven days as specified in the following schedule:

Monday	1300	Place in cabinet or room maintained at -10°F;
	1600	Remove from -10°F, and place in cabinet or room maintained at 120°F, 90 percent relative humidity and allow to remain overnight;
Tuesday	0800	Remove from 120°F, and place in 70°F;
	1300	Remove from 70°F and place in -10°F;
	1600	Remove from -10°F and place in 120°F (90 percent relative humidity);
Wednesday	0800	Remove from 120°F and place in 70°F;
	1300	Remove from 70°F and place in -10°F;
	1600	Remove from -10°F and place in 120°F (90 percent relative humidity);
Thursday	0800	Remove from 120°F and place in 70°F;
	1300	Remove from 70°F and place in -10°F;
	1600	Remove from -10°F and place in 120°F (90 percent relative humidity);

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Friday	0800	Remove from 120°F and place in 70°F;
	1300	Remove from 70°F and place in -10°F;
	1600	Remove one-half of cartridges and place in 120°F (90 percent relative humidity), retain balance at -10°F;
Saturday and Sunday		Maintain as at 1600 Friday;
Monday		Withdraw cartridges, condition at 70°F, subject to the ballistic test of 4.8.4.

4.8.5.2 Vibration. Cartridges from the production lot, selected in accordance with Table A2, Inspection Level III of MIL-STD-414 shall be subjected to the vibration test of MIL-D-21625 except that the temperature limits shall be +130°F to +135°F and -10°F to -15°F in lieu of that specified in MIL-D-21625. After vibration, the cartridges shall be subjected to the ballistic test of 4.8.4.

4.9 Criteria of acceptability.

4.9.1 Delay time. Cartridges shall be acceptable if they meet the requirements of 3.9.1 at the AQL values given in Table 1.

TABLE 1

Firing Temperature	Other Environmental Treatment	AQL
-15°F to -10°F	None	0.25
+70°F ± 2°F	None	0.25
+70°F ± 2°F	Temperature and Humidity Cycling Vibration	0.65
+70°F ± 2°F		0.65
+130°F to +135°F	None	0.25

4.9.2 Maximum pressure. The maximum pressure mean, variance, and standard deviation for each group of cartridges shall be computed separately. Cartridges subjected to temperature conditioning only shall be acceptable if the requirements of 3.9.2 are satisfied at an AQL of 0.65. The AQL value applicable to output of the cartridges from the environmental treatments shall be 1.0.

4.9.3 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

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4.10 Retest. There shall be no retests.

4.10.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and the complete test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with Bureau of Naval Weapons LD 537902.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 71-78.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with Bureau of Naval Weapons LD 537903.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

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5.3.2 Normal markings. In addition to the markings required by contract or order, and as required by Bureau of Naval Weapons LD 537902 and LD 537903, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. This MARK 87 MOD 0 delay cartridge was originally designed to be used in the release assembly of the TERRIER missile for booster separation.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required, and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List) which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Ordnance Systems Command and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Levels A and C packaging and packing given below were extracted from BUSANDA Instruction 4030.17.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

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6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirement of this level.

6.5 Explosives safety precautions.

6.5.1 Applicability. The loading, assembly and handling of the explosives, subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosive safety precautions. At plants of the Naval Ordnance Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Other warnings and precautions, pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching or firing), are included in the detailed technical requirements of the specification. Use of this specification shall not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.5.2 Personnel and equipment. Proper care must be exercised at all times to protect personnel from accidents, fires or explosions and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also paragraph 6.5.4, Static and electrical and other applicable parts of these safety precautions.)

6.5.2.1 Explosives operations shall be kept under strict supervision. Periodic inspections shall be employed to assure compliance with established safety regulations.

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6.5.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

6.5.2.3 Where required, remote control-barricaded-handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.2.4 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters. conveyors, and the like in safe operating condition.

6.5.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.2.7 During static tests, propulsion units such as rocket motors shall be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means shall be provided to harmlessly vent flames and exhaust gases.

6.5.3 Operating procedures. All loading operations shall be conducted in full accordance with a formalized operating procedure which shall include the following:

6.5.3.1 Keep only minimum, limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.3.2 Allow no fires, exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.3.3 Appropriate safety instructions for specific operations shall be posted and personnel instructed as to the hazards present and precautions involved.

6.5.3.4 The number of working personnel and transients permitted in each work space shall be established and posted in such areas. The numbers shall be the minimum number required to accomplish necessary operations.

6.5.3.5 All loading operations should be conducted in a neat and orderly manner.

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6.5.3.6 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings.

6.5.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.5.3.8 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.3.9 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See paragraph 6.5.4.)

6.5.4 Static and electrical.

6.5.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics, such as silk and nylon, which generate static electricity shall be avoided. Additional grounding devices, such as grounded bracelets for workers, should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type of electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations, such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device.

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Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast; also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects, but without inducing excessive moisture absorption by any of the components of the item being processed.

6.5.4.2 Protect all explosive operations from the effects of electric current originating in equipment (such as soldering irons or guns, heaters, switches, wiring, motors, lights, test instruments, etc.) by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric types of primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips or other devices should be employed during manufacturing operations involving items such as electric primers, detonators or squibs.

6.5.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5.5 Operating facility and storage regulations.

6.5.5.1 All explosives including partially loaded items should be stored in suitable storage magazines located in accordance with the minimum distances prescribed in the American Table of Distances (ATD). While the explosive or loaded items are in process, stored in loading rooms, or in ready or service magazines, they should be located in accordance with Intraplant Distances set forth in the ATD. For Navy managed or contractor managed-Navy owned-explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy and the Naval Ordnance Systems Command covering quantity-distance regulations for explosives apply.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81330A(OS) CARTRIDGE, DELAY, MARK 87 MOD 0

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

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